DIPLOMA PROGRAMME PRACTICE · SL · SET A

Chemistry / Paper 2

Worked mark scheme · 50 marks · 90 minutes

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each.

PAPER 2 · QUESTION 1 [10]

A salt that loses water

(a) Determine x, showing the mole ratio. [3]

1 Mass water=1.798 g, corresponding to 1.798/18.02=0.09978 mol.

1 Dry salt amount=3.192/159.6=0.02000 mol.

1 Ratio H_2O:$CuSO_4$=4.99:1, so x=5.

(b) Describe how to establish constant mass experimentally, including a precaution specific to this residue. [2]

1 Heat, cool in a desiccator and weigh, then repeat heating/cooling/weighing.

1 Stop when consecutive cooled masses agree within the balance precision; protection from moist air limits rehydration.

(c) All the dry residue is dissolved and made to 250.0 cm³. Find the Cu^{2+} concentration and the number of sulfate ions in 10.00 cm³ of this solution. Use NA=6.02×10²³ mol⁻¹. [3]

1 $[Cu^{2+}]$=0.02000/0.2500=0.08000 mol dm^{-3}.

1 $n(SO_4^{2-})$ in 10.00 cm^3=0.08000×0.01000=8.000×10^{-4} mol.

1 Number=4.82×10^{20} sulfate ions.

(d) Explain how splattering a little solid out of the crucible during heating would bias x if it were unnoticed. [2]

1 The lost solid would be assigned to water loss, overestimating water amount.

1 The measured dry-salt amount is too small, so the ratio and calculated x are too large.

PAPER 2 · QUESTION 2 [10]

Selecting a conductor

(a) Calculate the relative atomic mass. State the numbers of protons, neutrons and electrons in $^{65}Cu^{2+}$. [3]

1 Ar=0.700×63+0.300×65=63.60.

1 29 protons and 36 neutrons.

1 27 electrons.

(b) Write noble-gas configurations for neutral Cu and Cu^{2+}. [2]

1 Cu: [Ar]$3d^{10}4s^{1}$.

1 Cu^{2+}: [Ar]$3d^{9}$; the 4s electron is removed before a 3d electron.

(c) Explain why a copper wire conducts and can be drawn into a thinner wire without losing its metallic bonding. [3]

1 Positive metal ion cores are attracted to delocalized electrons.

1 Mobile electrons carry electrical charge.

1 Layers of cores can slip while non-directional attraction to the electrons persists.

(d) The supplier claims that isotope enrichment will greatly strengthen the wire by disrupting the lattice in the same way as adding a different metal. Evaluate the claim. [2]

1 Isotopes have the same nuclear charge and essentially the same electron structure and atomic size.

1 Unlike different-sized alloy atoms, isotope substitution does not cause comparable size distortion; the proposed mechanism does not justify the claim.

PAPER 2 · QUESTION 3 [10]

Two three-carbon feedstocks

(a) Identify which feedstock undergoes addition polymerization and draw two consecutive repeat units using a condensed formula with continuation bonds. [3]

1 Propene contains C=C and can undergo addition polymerization.

1 Two repeats: $–CH_2–CH(CH_3)–CH_2–CH(CH_3)–$ with continuation bonds.

1 The backbone bonds are single and the methyl branches are attached to alternating backbone carbons.

(b) Give a chemical test distinguishing the two feedstocks. Include the observation for each and state a condition that prevents confusion with a substitution reaction. [3]

1 Use bromine water without UV illumination.

1 Propene decolourizes it by addition across C=C.

1 Propane shows no rapid decolourization under these conditions; avoid UV initiation.

(c) Write the complete combustion equation for the saturated feedstock and find the mass of CO_2 released in the fuel test. [2]

1 $C_3H_8+5O_2\rightarrow3CO_2+4H_2O$.

1 0.300 mol CO_2 corresponds to 13.2 g.

(d) Explain why the polymer can insulate a wire while the metal core conducts. [2]

1 The covalent polymer lacks mobile ions or delocalized electrons available to move through it.

1 The metal contains mobile delocalized electrons, so the two materials have different electrical roles.

PAPER 2 · QUESTION 4 [10]

An energy estimate for a fuel cell reactant

(a) Use the bond data to estimate the combustion enthalpy to gaseous water. [3]

1 Breaking reactant bonds requires $436+\frac{1}{2}(498)=685$ kJ mol^{-1}.

1 Forming two O–H bonds releases $2\times463=926$ kJ mol^{-1}.

1 $\Delta H\approx685-926=-241$ kJ mol^{-1}.

(b) Calculate the enthalpy to liquid water. Explain why specifying the product state matters. [2]

1 Add condensation: $-241-44=-285$ kJ mol^{-1}.

1 Further heat is released when intermolecular attractions form in the liquid.

(c) A device converts 55.0% of the estimated energy for liquid-water formation into useful electrical output. Find the output for 4.00 g H_2. Use $M(H_2)=2.016$ g mol^{-1}. [3]

1 Amount=4.00/2.016=1.984 mol.

1 Estimated chemical energy=1.984×285=565 kJ.

1 Useful electrical energy≈311 kJ.

(d) Give one limitation of the bond-enthalpy estimate and one reason why the point-of-use CO_2 emission alone cannot establish the fuel's total carbon footprint. [2]

1 Mean bond enthalpies are averages over gaseous compounds; actual bond energies may differ.

1 Hydrogen production, compression or transport can consume fossil energy, which must be included in a life-cycle comparison.

PAPER 2 · QUESTION 5 [10]

Reconciling rate and yield

(a) Write the equilibrium expression for the forward reaction. Use the reverse-reaction constant to determine the forward constant. [3]

1 $K_c=[SO_3]^2/([SO_2]^2[O_2])$.

1 At fixed temperature, reversing the equation inverts the equilibrium expression: Kforward = 1/Kreverse. Also accept valid substitution of the supplied equilibrium concentrations.

1 $K_c=40.0$.

(b) The vessel is compressed at constant temperature. Predict the equilibrium change in amount of SO_3 and the change, if any, in Kc. [2]

1 Compression favours the side with fewer moles of gas (2 rather than 3), increasing amount of SO_3.

1 Kc remains unchanged at the same temperature.

(c) Explain why a catalyst can increase hourly production in a flow reactor without changing the equilibrium yield in a sealed vessel. [3]

1 It provides an alternative pathway with lower activation energy.

1 It speeds both forward and reverse reactions and shortens the time needed to approach equilibrium.

1 It does not change equilibrium composition or K at fixed temperature.

(d) A manager proposes increasing temperature to maximize both rate and equilibrium yield. Evaluate both parts of the proposal. [2]

1 A larger fraction of collisions can overcome the activation barrier, generally increasing rate.

1 Higher temperature favours the endothermic reverse direction, reducing equilibrium yield of SO_3; the aims conflict.